Prime Stack Defect Report

Course: Software Validation, Verification and Security Testing
Application under test: Frost Inventory and Shopping System
Team: Prime Stack
Environment: Local backend `http://localhost:3000`, frontend `http://localhost:4200`
Report date: 19 May 2026

1. Defect Origin Summary

Not every defect came from the same test type. DEF-001 and DEF-002 were confirmed by backend functional tests. DEF-005, DEF-006, and DEF-007 come from security test design/scripted security checks. DEF-008 comes from the ZAP scan/header verification. DEF-003 and DEF-004 are source-code review/API robustness defects. DEF-009 and DEF-010 were confirmed by expanded Playwright profile coverage. DEF-011 and DEF-012 come from teammate manual usability review.

ID	Title	Origin	Severity	Status
DEF-001	Missing/invalid email returns 500 instead of 400	Functional tests TC-FUNC-002, TC-FUNC-008	Medium	Fixed
DEF-002	Template literal bug in `getPurchaseByUserId`	Functional test TC-FUNC-032	Medium	Fixed
DEF-003	`SupplierRoute` POST has no error handling	Source-code review	Medium	Fixed
DEF-004	Login route hardcodes status 500	Functional/API error-handling review	Medium	Fixed
DEF-005	Mass assignment allows any user to register as admin	Security test ST-005	Critical	Fixed
DEF-006	`/User/users` exposes all user data without authentication	Security test ST-007 and route review	High	Fixed
DEF-007	XSS payload stored in database without sanitization	Security test ST-003	Medium	Fixed
DEF-008	Missing security headers	ZAP scan/header verification	Low	Fixed
DEF-009	Login stores value objects in session, causing `[object Object]` in UI	Playwright profile test UT-007	Medium	Fixed
DEF-010	Profile refresh assigns wrapped API response as user state	Playwright profile edit test UT-009	Medium	Fixed
DEF-011	Inconsistent background color on registration input fields	Manual usability test UT-MAN-001	Low	Open
DEF-012	Missing search and filtering functionality for product discovery	Manual usability test UT-MAN-002	Medium	Open

2. Functional Test Evidence Summary

Before the fixes, the functional suite result was:

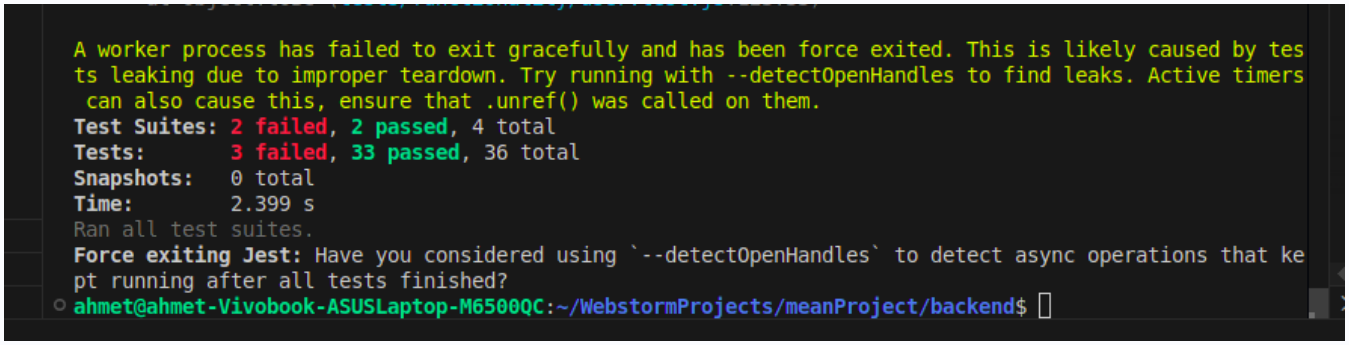
```text
Test Suites: 2 failed, 2 passed, 4 total
Tests: 3 failed, 33 passed, 36 total
```

After fixing the functional and API defects, the functional suite result was:

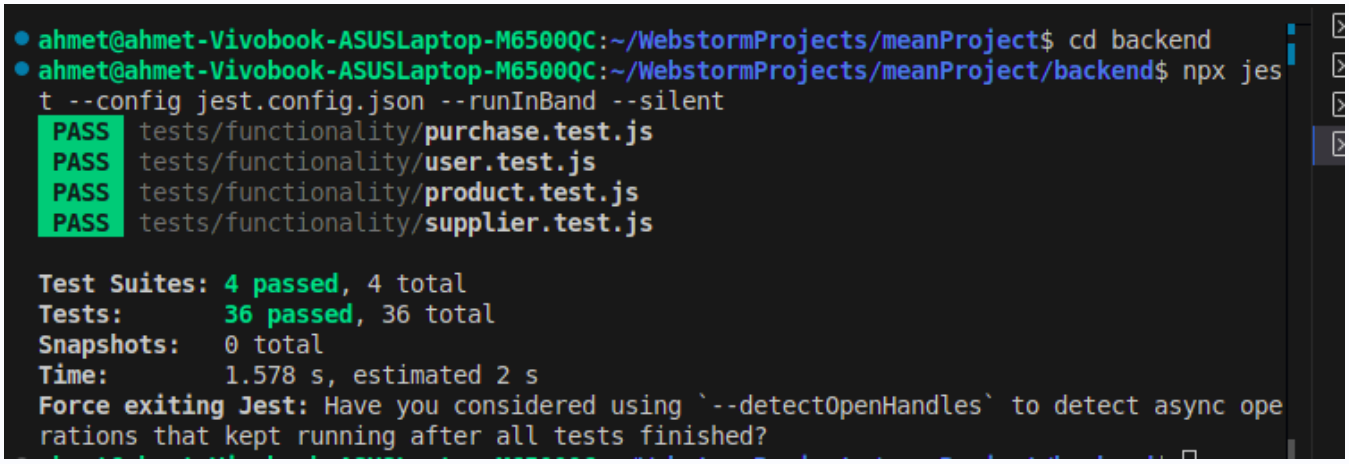
```text
Test Suites: 4 passed, 4 total
Tests: 36 passed, 36 total
```

After fixing the security defects, the k6 security result was:

```text
checks_succeeded: 100.00% 8 out of 8
checks_failed: 0.00% 0 out of 8
```



Functional test result before fixes



Functional test result after fixes

DEF-001: Missing/Invalid Email Returns 500 Instead of 400

Origin

Functional testing. Confirmed by:

- `TC-FUNC-002: should reject registration with missing email`
- `TC-FUNC-008: should reject invalid email format`

Details

Field	Value
Severity	Medium
Status	Fixed
Test file	`backend/tests/functionality/user.test.js`
Source files	`backend/domain/user/valueObjects/Email.js`, `backend/api/UserRoute.js`

Before the fix, `Email.js` threw plain `Error` objects. `UserRoute.js` checks `error.statusCode`; plain errors do not have that property, so the route fell back to `500`.

Before:

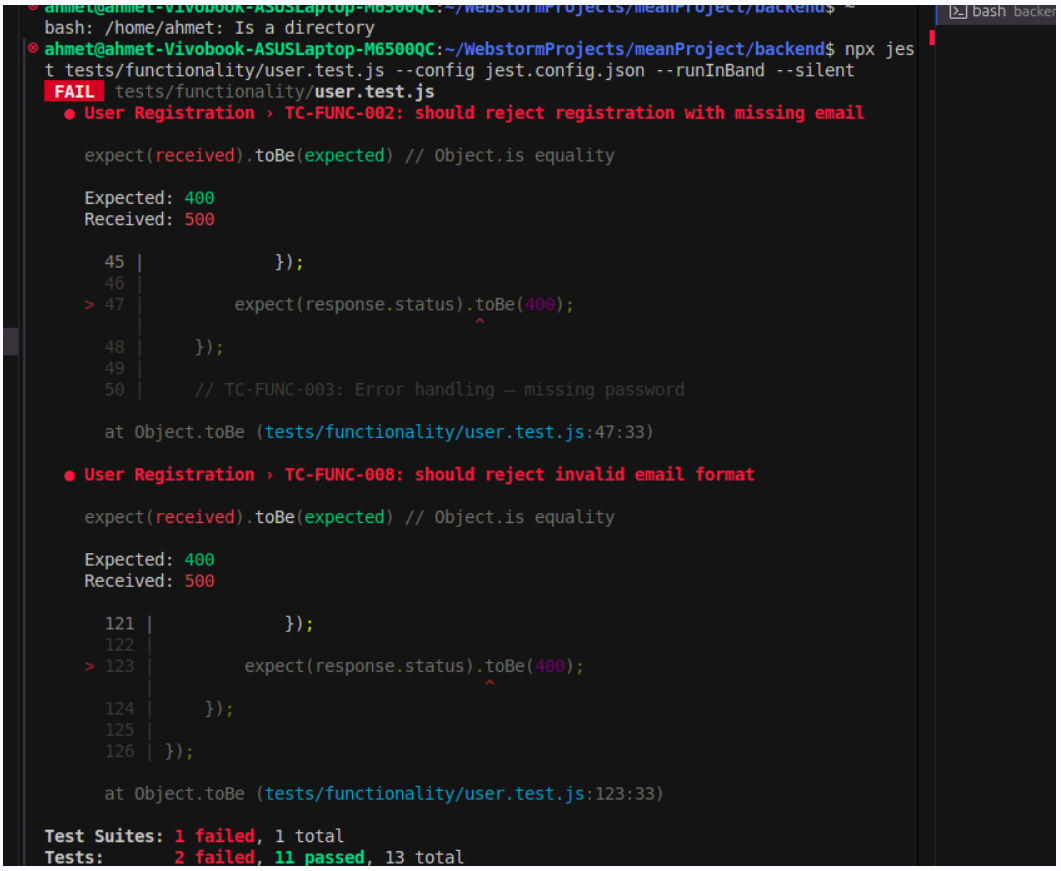
```js
throw new Error("Email required");
throw new Error("Invalid email format");
```

After:

```js
throw new ValidationError("Email required");
throw new ValidationError("Invalid email format");
```

The route behavior then correctly maps the validation failure to `400 Bad Request`.

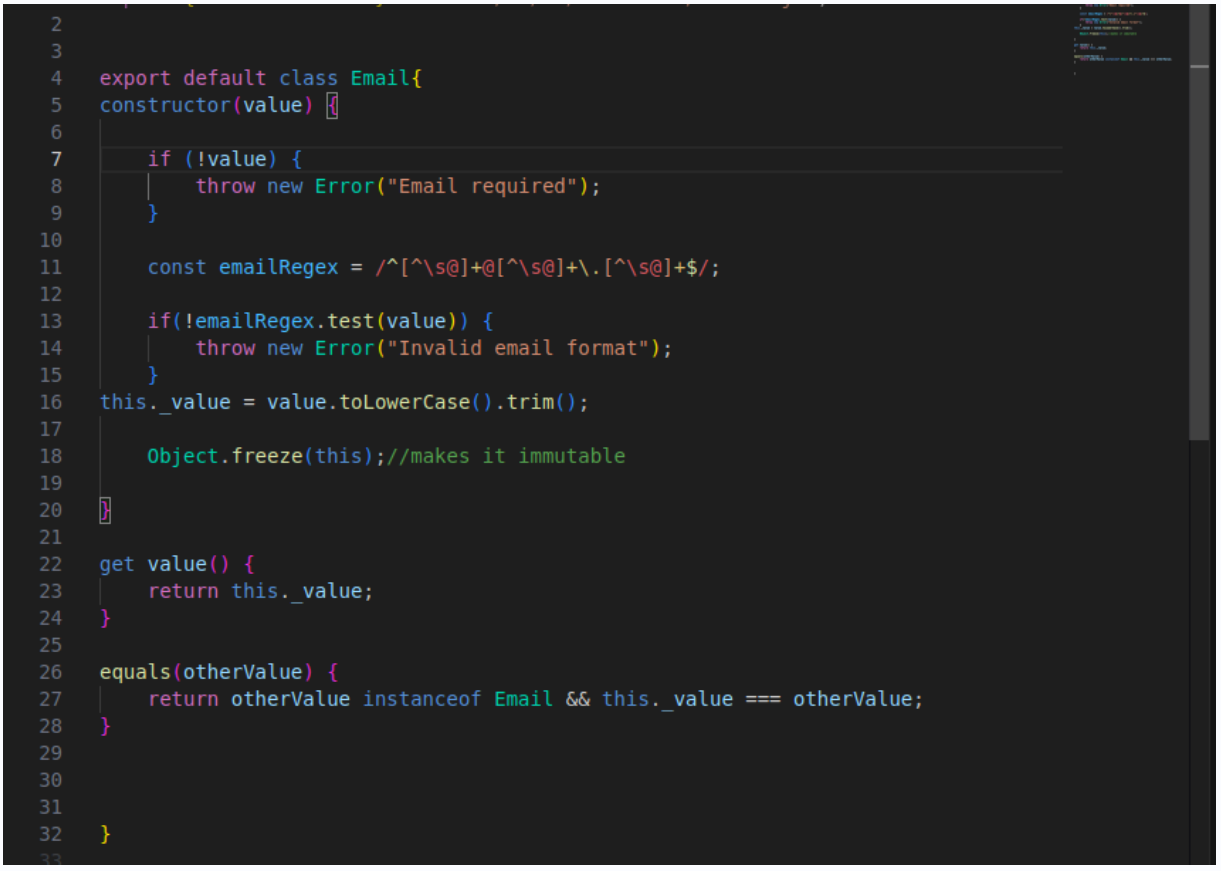
Evidence



DEF-001 failing user tests

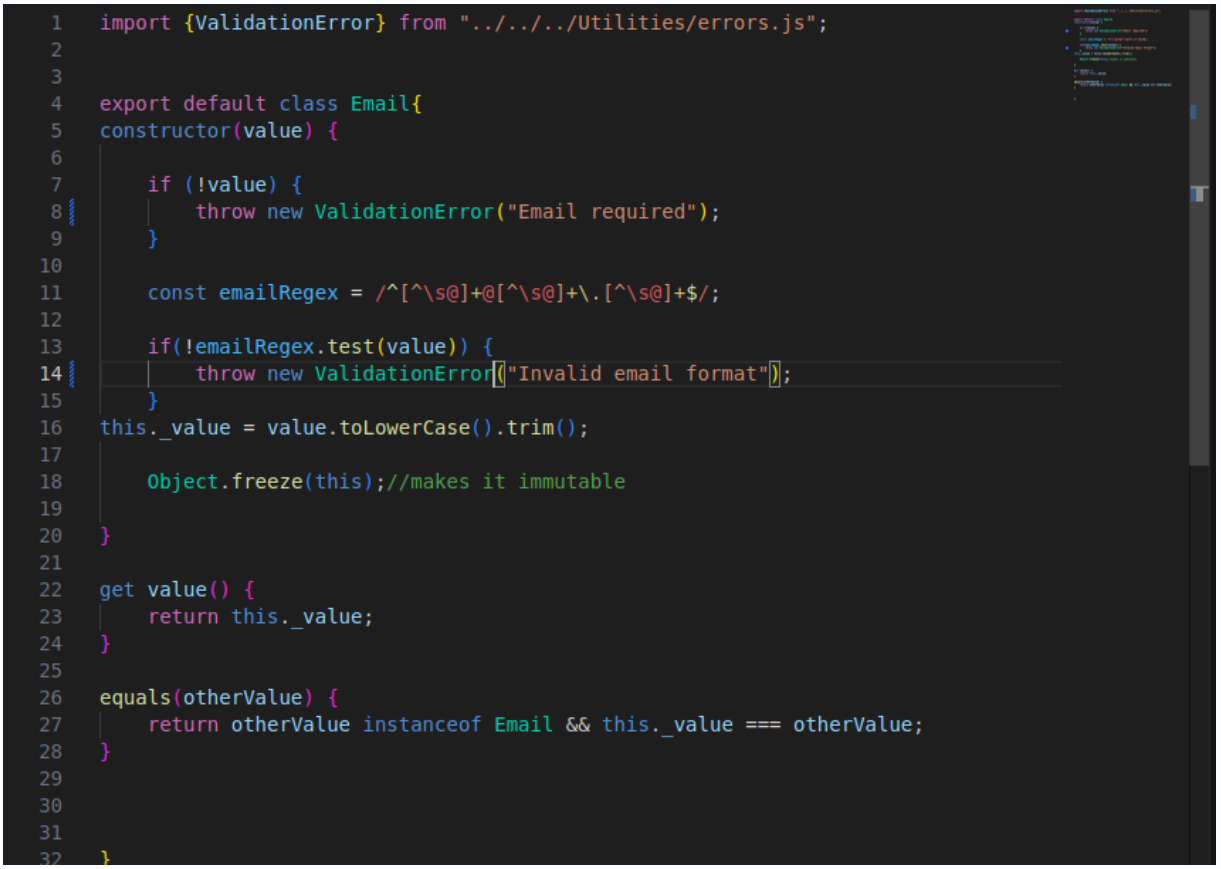
```
    }
});


router.post("/register", async (req, res) => {
    try {

        const user = req.body;
        const result = await userApplication.createUser(user);

        res.status(201).json({
            message: 'User registered successfully',
            user: result.toSafeObject()
        });


    } catch (error) {
        console.error('❌ Registration error:', error);
        const statusCode = error.statusCode || 500;

        res.status(statusCode).json({
            error: 'Registration failed',
            message: error.message
        });

    }
});

router.post("/login", async (req, res) => {

    try {
```

DEF-001 route fallback behavior



```


export default class Email{
constructor(value) {

    if (!value) {
        throw new Error("Email required");
    }

    const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if(!emailRegex.test(value)) {
        throw new Error("Invalid email format");
    }
this._value = value.toLowerCase().trim();

    Object.freeze(this);//makes it immutable

}

get value() {
    return this._value;
}

equals(otherValue) {
    return otherValue instanceof Email && this._value === otherValue;
}



}
```

DEF-001 before fix: plain Error in Email value object



```
import {ValidationError} from "../../../Utilities/errors.js";


export default class Email{
constructor(value) {

    if (!value) {
        throw new ValidationError("Email required");
    }

    const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if(!emailRegex.test(value)) {
        throw new ValidationError("Invalid email format");
    }
this._value = value.toLowerCase().trim();

    Object.freeze(this);//makes it immutable

}

get value() {
    return this._value;
}

equals(otherValue) {
    return otherValue instanceof Email && this._value === otherValue;
}



}
```

DEF-001 after fix: ValidationError in Email value object

DEF-002: Template Literal Bug in `getPurchaseByUserId`

Origin

Functional testing. Confirmed by:

- `TC-FUNC-032: should return purchases for a valid user id`

Details

Field	Value
Severity	Medium
Status	Fixed
Test file	`backend/tests/functionality/purchase.test.js`
Source file	`backend/domain/purchase/PurchaseRepository.js`

Before the fix, the SQL string used normal quotes, so `${orderClause}` was sent to MySQL literally.

Before:

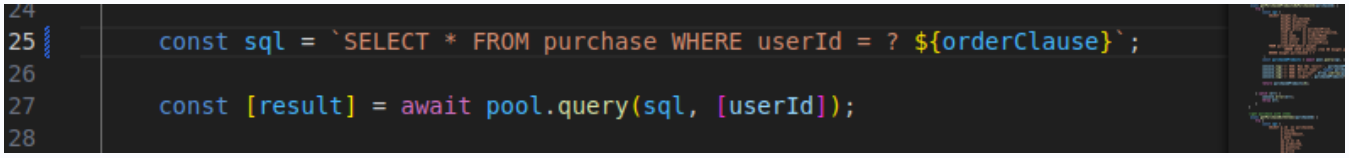
```js
const sql = 'SELECT * FROM purchase WHERE userId = ? ${orderClause}';
```

After:

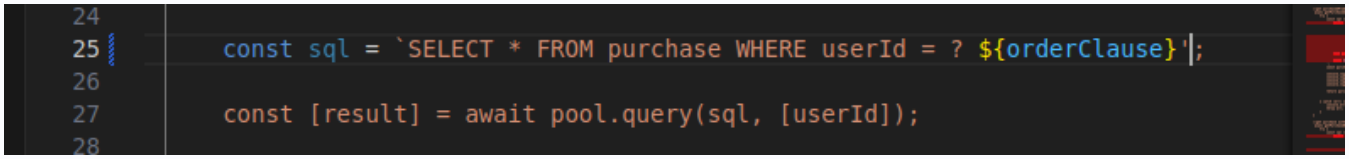
```js
const sql = `SELECT * FROM purchase WHERE userId = ? ${orderClause}`;
```

This allows `ORDER BY date ASC` or `ORDER BY date DESC` to be inserted into the SQL query correctly.

Evidence



DEF-002 before fix: order clause not interpolated



DEF-002 after fix: template literal used

DEF-003: `SupplierRoute` POST Has No Error Handling

Origin

Source-code review.

Details

`POST /Supplier/` directly awaits supplier creation without local `try/catch` handling. If validation or duplicate checks fail, the API may return an uncontrolled server error instead of a structured `400` or `409` response.

Fix applied: supplier creation is now wrapped in `try/catch`. The route uses `error.statusCode || 500`, returns a structured JSON error body, and still preserves the previous success behavior.

DEF-004: Login Route Hardcodes Status 500

Origin

Functional/API error-handling review.

Details

The login route returns `500` for invalid login cases. Wrong password, unknown username, and missing credentials should be client/authentication failures.

Expected statuses:

- Missing username/password: `400 Bad Request`
- Wrong username/password: `401 Unauthorized`
- Unexpected backend failure: `500 Internal Server Error`

Fix applied: the login route now uses `error.statusCode || 500`, the same pattern used by the registration route.

Verification:

- Wrong username/password returns `401 Unauthorized`.
- Missing credentials returns `400 Bad Request`.
- Functional suite passed with `36/36` tests.

Evidence

```
    } catch (error) {
        console.error('❌ Login error:', error);
-       res.status(500).json({
+       const statusCode = error.statusCode || 500;
+
+       res.status(statusCode).json({
            error: '  Login failed',
            message: error.message
        });
:
```

DEF-004 login route status-code fix

DEF-005: Mass Assignment Allows Any User to Register as Admin

Origin

Security test ST-005.

Details

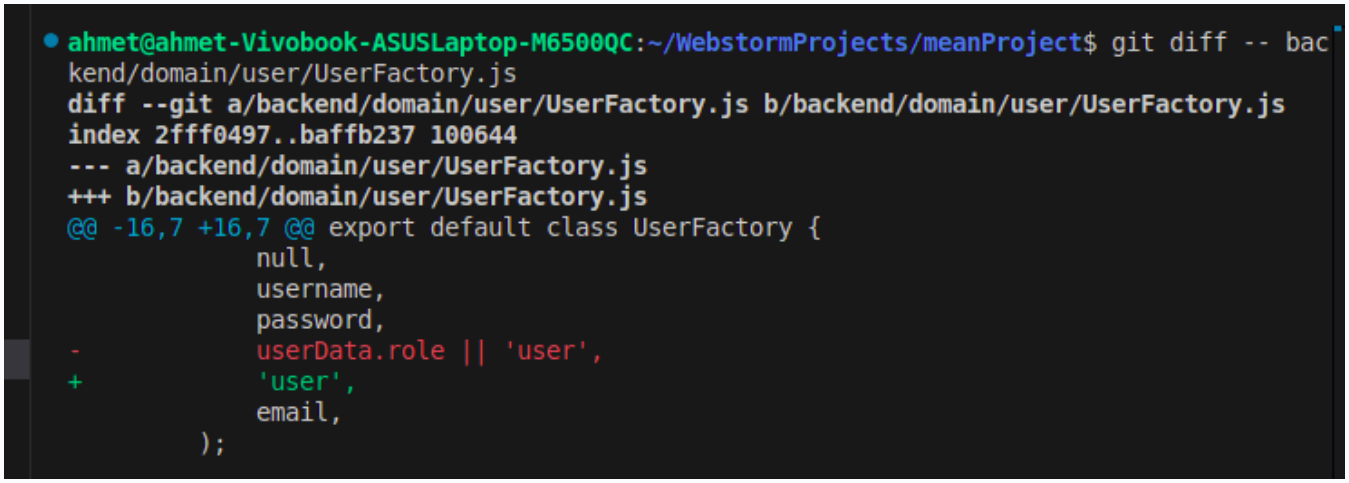
The security test attempts to register with `role: "admin"` in the request body. Public registration should not accept privileged fields from the client.

Fix applied: public registration no longer trusts `userData.role`. `UserFactory.createUser()` always creates normal users with role `user`; privileged admin creation must happen through the separate `createAdmin()` path.

Verification from k6:

```text
ST-005 Mass Assignment: 201 ... "role":"user"
✓ ST-005 mass assignment does not grant admin role
```

Evidence



DEF-005 mass-assignment role fix

DEF-006: `/User/users` Exposes All User Data Without Authentication

Origin

Security test ST-007 and route review.

Details

`GET /User/users` should require authentication and admin authorization. Public access can expose user data and support account enumeration.

Expected behavior:

- Unauthenticated request: `401 Unauthorized`
- Authenticated non-admin request: `403 Forbidden`
- Admin request: safe user list only

Fix applied: `/User/users` now uses `requireAdmin`, which returns `401` for unauthenticated users and `403` for users without sufficient privileges.

Verification from k6:

```text
ST-007 Unauth User List: 401 - {"error":"Authentication required"}
✓ ST-007 user list requires authentication
```

Evidence

```
@@ -177,10 +179,10 @@ router.get('/profile', requireAuth, async (req, res) => {
 });

 //get all users
-router.get('/users', async (req, res) => {
+router.get('/users', requireAdmin, async (req, res) => {
     try {
         const allUsers = await userApplication.getAllUsers();
-        res.status(201).json(allUsers)
+        res.status(200).json(allUsers)
     } catch (error) {
         console.error("Failed to fetch all users", error);
         res.status(500).json({
@@ -219,4 +221,4 @@ router.put('/', async (req, res) => {
 });
```

DEF-006 user list admin authorization fix

DEF-007: XSS Payload Stored in Database Without Sanitization

Origin

Security test ST-003.

Details

The security test sends `<script>alert("xss")</script>` as a username. The application should reject or sanitize this input before storage/display.

Fix applied: the `Username` value object now rejects `<` and `>` characters, preventing script-tag usernames from being stored.

Verification from k6:

```text
ST-003 XSS Registration: 400 - {"error":"Registration failed","message":"Username contains invalid characters"}
✓ ST-003 XSS payload in username is handled
```

Evidence

```
+++ b/backend/domain/user/valueObjects/Username.js
@@ -11,6 +11,10 @@ export default class Username {
             throw new ValidationError("Username must be between 3-50 characters");
         }

+        if (/[<>]/.test(value)) {
+            throw new ValidationError("Username contains invalid characters");
+        }
+
         this._value = value.trim();
         Object.freeze(this);
     }
@@ -20,4 +24,4 @@ export default class Username {
     }
```

DEF-007 username validation fix

DEF-008: Missing Security Headers

Origin

ZAP scan.

Details

ZAP reported missing or weak security headers:

Header issue	Risk	Evidence
CSP missing no-fallback directives	Medium	`Content-Security-Policy: default-src 'none'` without `frame-ancestors` and `form-action`
`X-Content-Type-Options` missing	Low	`/Product/` response does not include `nosniff`
`X-Powered-By` exposed	Low	Response contains `X-Powered-By: Express`

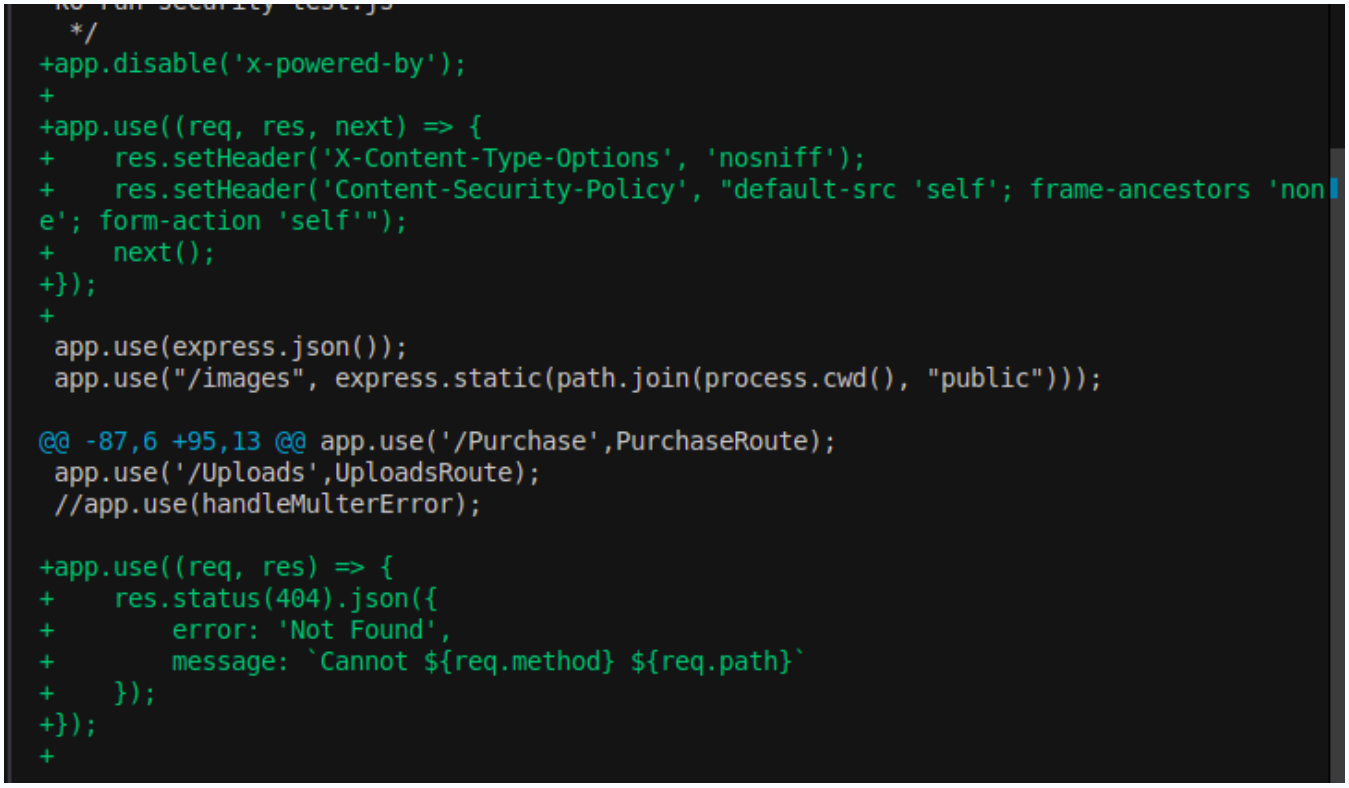
Fix applied:

- `app.disable('x-powered-by')`
- `X-Content-Type-Options: nosniff`
- `Content-Security-Policy: default-src 'self'; frame-ancestors 'none'; form-action 'self'`
- Custom JSON `404` handler so missing routes such as `/robots.txt` no longer fall back to Express's default CSP.

Verification from header check:

```text
HTTP/1.1 404 Not Found
X-Content-Type-Options: nosniff
Content-Security-Policy: default-src 'self'; frame-ancestors 'none'; form-action 'self'
```

Evidence



DEF-008 security header middleware fix

DEF-009: Login Stores Value Objects in Session, Causing `[object Object]` in UI

Origin

Frontend Playwright profile testing. Confirmed by:

- `UT-007: Client can view profile and open edit profile form`

Details

Field	Value
Severity	Medium
Status	Fixed
Test file	`frontend/tests/example.spec.ts`
Source file	`backend/api/UserRoute.js`

The login route stored `user.username` and `user.email` directly in the session. In some cases those fields are value objects, so `/User/status` returned an object instead of a string. The Angular profile screen then rendered `Welcome, [object Object]!`.

Fix applied: the login route now stores primitive string values using `.value` or `._value` before falling back to the raw field.

Verification:

```text
PASS [chromium] UT-007: Client can view profile and open edit profile form
PASS [firefox] UT-007: Client can view profile and open edit profile form
```

DEF-010: Profile Refresh Assigns Wrapped API Response as User State

Origin

Frontend Playwright profile edit testing. Confirmed by:

- `UT-009: Client can edit profile username and email`

Details

Field	Value
Severity	Medium
Status	Fixed
Test file	`frontend/tests/example.spec.ts`
Source file	`frontend/src/app/components/clientPanels/client-profile/client-profile.ts`

After a profile update, the profile component refreshed the user profile but assigned the full API response wrapper to `currentUser`. Because `/User/profile` returns `{ message, user }`, the page could render blank profile fields after saving.

Fix applied: `fetchFreshProfile()` now assigns `response.user || response`, so both wrapped and direct user responses are handled.

Verification:

```text
PASS [chromium] UT-009: Client can edit profile username and email
PASS [firefox] UT-009: Client can edit profile username and email
```

DEF-011: Inconsistent Background Color on Registration Input Fields

Origin

Manual usability testing from teammate review. Related observation:

- `UT-MAN-001: Registration visual consistency`

Details

Field	Value
Severity	Low
Priority	Medium
Status	Open
Source	Manual usability test
Area	Frontend `/login` registration form

On the registration form, the username and password inputs appear with a gray background while the email input appears white. Test users may interpret the gray fields as disabled or less editable, which slows down the registration flow.

Steps to reproduce:

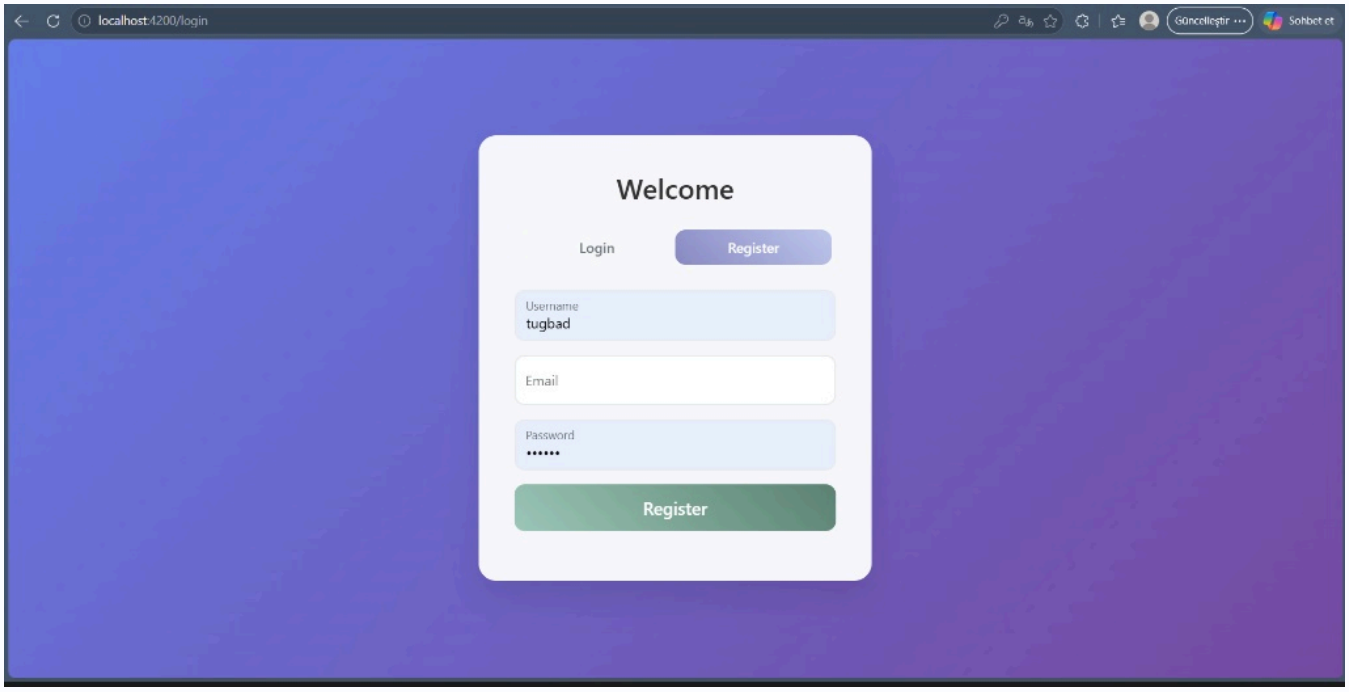
1. Navigate to `http://localhost:4200/login`.
2. Click the `Register` tab.
3. Compare the username, email, and password input field backgrounds.

Expected result: all active registration inputs use consistent visual styling.

Actual result: input field background colors are inconsistent.

Suggested fix: unify CSS styling for all input fields in the login/register component.

Evidence



DEF-011 registration input color evidence

DEF-012: Missing Search and Filtering Functionality for Product Discovery

Origin

Manual usability testing from teammate review. Related observation:

- `UT-MAN-002: Product discovery/search`

Details

Field	Value
Severity	Medium
Priority	High
Status	Open
Source	Manual usability test
Area	Frontend client dashboard / product discovery

The client dashboard and product discovery workflow do not provide search or category filtering. As inventory grows, users may need to scan manually through product cards, increasing task time.

Steps to reproduce:

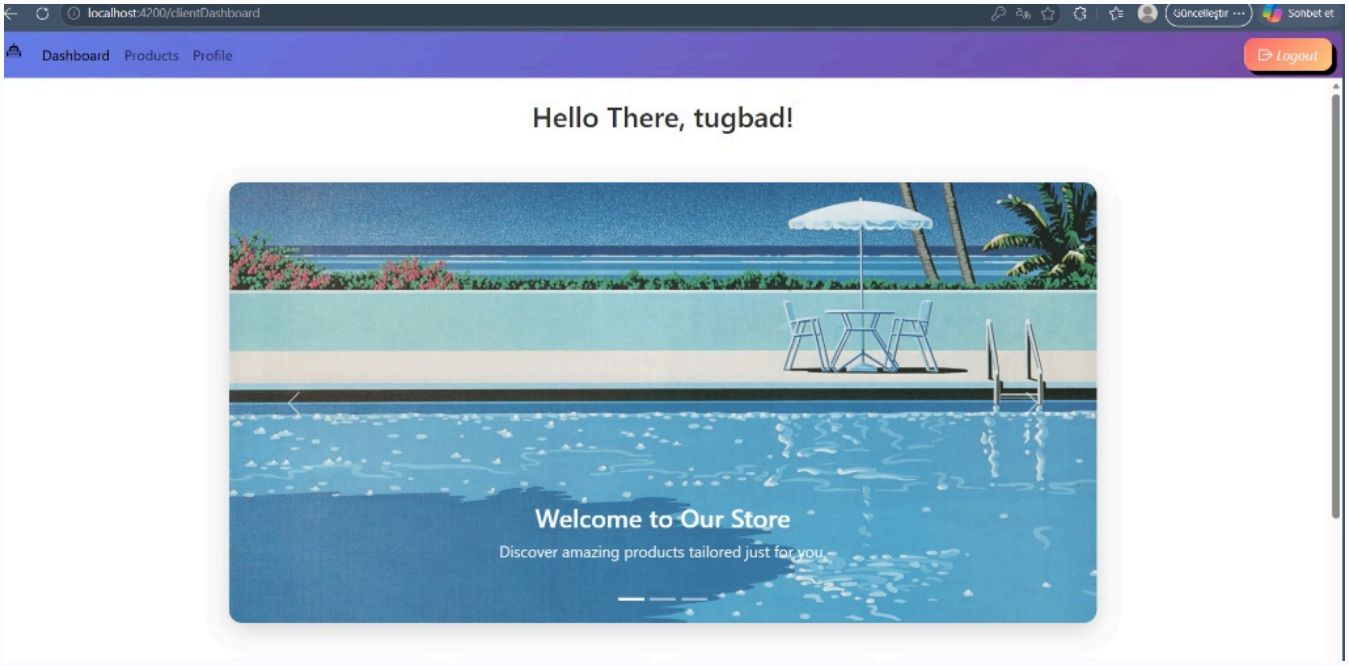
1. Log in as a client user.
2. Navigate to the client dashboard and product browsing workflow.
3. Attempt to search for a specific product name or filter by category/supplier.

Expected result: a search bar and basic filtering controls are available.

Actual result: no search or filtering functionality is available in the observed dashboard/product discovery flow.

Suggested fix: add product search and filter controls to the client-facing product discovery workflow.

Evidence



DEF-012 missing search/filter evidence